Titus Ammerman

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PROFESSIONAL SUMMARY

M.S. Applied Statistics graduate specializing in geospatial modeling, spatiotemporal forecasting, and remote sensing. U.S. Air Force veteran with previous Secret clearance, FAA Part 107 certification, and applied experience spanning industrial IoT systems, satellite imagery analysis, and NSF-funded spatial research.

EDUCATION

M.S. Applied Statistics, Data Science Concentration **Aug 2024 – June 2026**
DePaul University

Graduate Certificate in Geographic Information Systems (GIS) **Jan 2025 – March 2026**
DePaul University

B.A. Data Science **Aug 2021 - June 2024**
Franklin College

EXPERIENCE

Graduate Research Assistant **Jan 2026 – June 2026**
DePaul University

- Developed Python scripts to automate standardization and validation of open survey responses for an NSF-funded ($750k+) study on equitable public space access, reducing manual review burden on the research team
- Implemented string manipulation and rule-based flagging logic to correct geocodeable location entries and filter invalid responses (e.g. non-public or unspecified locations), enabling downstream spatial analysis
- Applied a two-digit validation coding system to categorize ambiguous entries for secondary researcher review, supporting data integrity across 1,500+ survey records

Industrial Data Science Intern **Apr 2023 – Sep 2023**
WPR Services, Indiana IoT Laboratory

- Architected industrial data acquisition systems, utilizing Modbus TCP/RTU protocols and hexadecimal register manipulation to digitize raw electrical telemetry for real-time monitoring
- Optimized data throughput and network reliability by configuring IPv4/v6 subnets and resolving serial communication bottlenecks (baud rate calibration) to ensure high-fidelity data transfer
- Developed a sensor-based box dimensioning prototype for high-speed conveyor belts; performed latency analysis and showcased the functional system at CorrExpo to demonstrate automated monitoring for the corrugated packaging industry

Electro-Environmental Systems Technician **Mar 2017 – Sept 2023**
United States Air Force *(Active: 2017–2021 | Reserve: 2021–2023)*

- Maintained life-cycle readiness for electrical and pneumatic systems across a KC-135 fleet, sustaining a 95%+ mission-capable rate in time-critical operational environments
- Diagnosed and resolved complex multi-system failures under operational pressure, where equipment downtime carried direct mission consequences
- Authored fleet-wide revisions to technical maintenance documentation, standardizing procedures across crews to reduce procedural error risk

Mathematics Tutor **Oct 2021 – Mar 2026**
DePaul University & Franklin College

- Delivered on-demand instructional support across business mathematics, statistics, discrete math, and introductory calculus, with additional coverage of linear algebra and multivariable calculus
- Adapted explanations in real time to diverse student needs and varying levels of preparation, with demand spiking significantly around midterms and finals

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PROJECTS

Urban Heat Island Analysis | R, ArcGIS Pro, Google Earth Engine, Google Cloud Platform **Mar 2026**

- Designed and executed a neighborhood-level geospatial risk analysis of urban heat exposure inequities across Chicago census tracts, integrating LANDSAT 9 thermal satellite imagery (30m resolution), CDC PLACES health outcome data, and U.S. Census demographic data to quantify heat burden and associated health risks
- Applied Local Indicators of Spatial Association (LISA) with Rook and Queen contiguity specifications to identify spatial clusters and hotspots of land-surface temperature, heat-related emergency room visits, cardiovascular risk, and general mortality across census tracts - demonstrating experience in geospatial hazard exposure analysis
- Compared OLS and Generalized Additive Model (GAM) specifications to capture non-linear heat-health relationships, validating model fit and quantifying uncertainty across spatial units

Spatiotemporal Population Forecasting | Python, GeoPandas, pmdarima **June 2024**

- Engineered a spatiotemporal forecasting pipeline on the Cornell Lab eBird dataset, discretizing North American species ranges into 1 degree x 1 degree quadrat grid cells and modeling each as an independent SARIMAX time series to capture strong migratory seasonality
- Automated model selection across 300–455 populated cells per species using pmdarima's auto_arima; achieved median MAE of 1.21–3.89 sightings per cell across three focal species (American Black Duck, Cerulean Warbler, Northern Saw-whet Owl)
- Conducted in collaboration with DePaul's Wildlife Biology department under faculty ornithologist oversight as undergraduate capstone research

TECHNICAL SKILLS & CERTIFICATIONS

FAA Part 107 Remote Pilot (UAS Operator) | Federal Aviation Administration (FAA) **Feb 2026**
U.S. Security Clearance: Secret (Previous | Reinstatement-Eligible)
Languages: English (Native), Spanish (Near Native/Bilingual)
Statistical Programming: R (rgee, Tidyverse, Shiny), Python (PySpark, pandas, NumPy, Scikit-learn), SQL (BigQuery, MSSQL), Julia, C#
Geospatial & Remote Sensing: ArcGIS Pro, Google Earth Engine (GEE), Landsat, MODIS
Big Data & Cloud Infrastructure: AWS EMR, Hadoop/Spark, AWS CLI, GCP CLI
Software & Developer Tools: Bash/Unix, Git, Excel (VBA/Power Query), Microsoft Access, .NET Framework